



Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan
College of Engineering
(An Autonomous Institution affiliated to
Rashtrasant Tukadoji Maharaj Nagpur
University)



Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

(Signature and Date in Handwritten)

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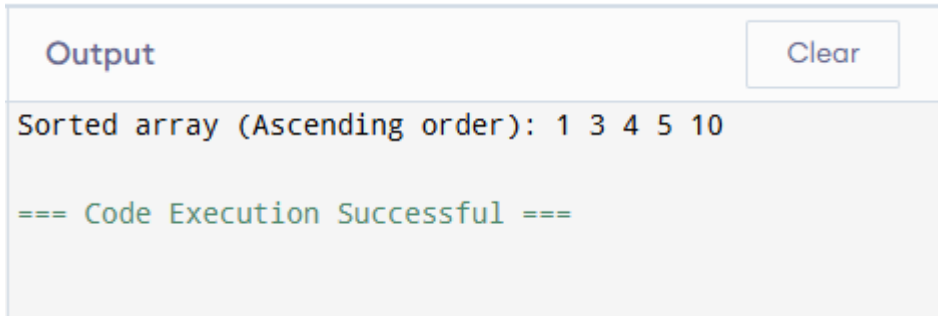
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Session	2024-25 (ODD)	Course Name	Computer vision Lab
Semester	4	Course Code	CT
Roll No	120	Name of Student	Shrusti Katakwar
Practical Number	02		
Course Outcome	1. Understand the fundamental concepts of non-linear data structures 2. Implement the real-life application using nonlinear data structures. 3. Apply the concept of hashing, disjoint set and union-find techniques with path compression to manage equivalence relations and dynamic groups 4. Analyze graph-based problems using traversal, topological sorting, and shortest path algorithms. 5. Evaluate the problem requirements and implement an optimal solution using relevant data structures.		
Aim	Construct a max/min heap and apply heap sort to arrange elements in ascending or descending order.		
Problem Definition	Construct a max/min heap and apply heap sort to arrange elements in ascending or descending order.		
Theory (100 words)	A Heap is a special type of complete binary tree that satisfies the heap property. A complete binary tree is a tree in which: All levels are completely filled except possibly the last level.		

		The last level is filled from left to right.	
		There are two types of heaps: - Max Heap - Min Heap	
		Types of Heap (a) Max Heap A Max Heap is a complete binary tree in which the value of each parent node is greater than or equal to the values of its children. - The largest element is always present at the root. - Used for sorting elements in ascending order using heap sort.	
		(b) Min Heap A Min Heap is a complete binary tree in which the value of each parent node is smaller than or equal to the values of its children. - The smallest element is always present at the root. - Used for sorting elements in descending order using heap sort. Construction of Heap Heap is generally constructed from an array. Steps for constructing a heap: - Represent the array as a complete binary tree. - Start from the last non-leaf node. - Compare the parent node with its children. - If the heap property is violated, swap the elements. - Repeat the process until the heap property is satisfied for all nodes. This process is called Heapify.	
		4. Heap Sort Heap Sort is a comparison-based sorting technique based on the heap data structure. It consists of two main phases: Phase 1: Build Heap	

	- Build a Max Heap for ascending order. - Build a Min Heap for descending order.	
	Phase 2: Sorting Process - Swap the root element with the last element. - Reduce the heap size. - Apply heapify to restore heap property. - Repeat the process until all elements are sorted.	
	5. Time and Space Complexity - Time Complexity: $O(n \log n)$ - Space Complexity: $O(1)$ - Heap Sort is an in-place sorting algorithm. Advantages of Heap Sort - Efficient for large data sets. - Does not require additional memory. - Guaranteed time complexity of $O(n \log n)$.	
Procedure and Execution	Algorithm:	
(100 Words)	Step 1: Start Step 2: Build Max Heap - For $i = (n/2 - 1)$ down to 0 - Apply HEAPIFY(arr, n, i) (This arranges the array into a Max Heap.)	
	Step 3: Perform Sorting - For $i = n - 1$ down to 1 - Swap arr[0] with arr[i] - Reduce heap size by 1 - Apply HEAPIFY(arr, i, 0) - Repeat until i becomes 1	
	Step 4: Stop Now the array is sorted in ascending order.	

		Code: <pre> #include <iostream> using namespace std; void heapify(int arr[], int n, int i) { int largest = i; int left = 2*i + 1; int right = 2*i + 2; if (left < n && arr[left] > arr[largest]) largest = left; if (right < n && arr[right] > arr[largest]) largest = right; if (largest != i) { swap(arr[i], arr[largest]); heapify(arr, n, largest); } } void heapSort(int arr[], int n) { for (int i = n/2 - 1; i >= 0; i--) heapify(arr, n, i); for (int i = n-1; i > 0; i--) { swap(arr[0], arr[i]); heapify(arr, i, 0); } } int main() { int arr[] = {4, 10, 3, 5, 1}; int n = sizeof(arr)/sizeof(arr[0]); heapSort(arr, n); cout << "Sorted array (Ascending order): "; for (int i = 0; i < n; i++) cout << arr[i] << " "; return 0; } </pre>	
		Output: 	
Output Analysis	Elements are arranged from smallest to largest. Example: If input = 4, 10, 3, 5, 1 Output = 1, 3, 4, 5, 10 This shows that Heap Sort successfully sorted the array in ascending order.		
Link of student Github profile where lab assignment has			



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Conclusion	❑ Heap is a complete binary tree that follows a special order rule. ❑ Max Heap keeps the largest element at the top. ❑ Min Heap keeps the smallest element at the top. ❑ Heap Sort uses heap structure to sort elements. ❑ It works efficiently with time complexity $O(n \log n)$. ❑ It does not require extra memory (in-place sorting).
Plag Report (Similarity index < 12%)	Plagiarism Report for 120.2.ads.pdf The attached file "120.2.ads.pdf" is a lab assignment on heap sort from Yeshwantrao Chavan College of Engineering, submitted by Shruti Katakwar (Roll No. 120). It includes standard theory on max/min heaps, construction, heap sort algorithm, code, output, and conclusion with repeated college headers and department visions. @ 120.2.ads.pdf Existing Plagiarism Note The document already contains a pre-filled "Plag Report" section stating "Similarity index 12%" dated 19/02/26. This appears to be a template placeholder for the required academic format rather than a generated analysis. @ 120.2.ads.pdf Analysis Limitations I cannot perform an external plagiarism scan or generate a new report, as no integrated tool accesses plagiarism databases like Turnitin or compares against the web. The content covers common topics (e.g., heap properties, $O(n \log n)$ complexity) likely matching textbooks or online tutorials, but originality cannot be verified without specialized software. For an official report under 12%, use free tools like SmallSEOTools or Grammarly Plagiarism Checker with the PDF upload. @ 120.2.ads.pdf
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